ELEC2134 : Circuits and Signals

PART A (Week 1 to Week 5)

Transform Methods

- Periodic signals and Fourier series
- Aperiodic signals and Fourier transforms
- Laplace transforms of signals and circuits

System Analysis

- Linear systems and convolution
- Impulse response
- Frequency response and transfer functions
- Steady-state analysis of linear systems
- Transient analysis of 1st and 2nd systems
- Circuit analysis using transform methods

PART B (Week 7 to Week 10)

AC Circuit Analysis

- AC circuits with sinusoidal inputs
- Use of phasors and complex impedance in AC circuit analysis
- AC power: real, reactive and apparent power
- Power factor (leading/lagging)
- Series and parallel resonance
- Transformers and coupled coils
- Application of network theorems: Two port networks

This week: turning $H(j\omega)$ into a practical graphical tool for understanding gain and phase across frequency.

Dr. Deepak Mishra, School of EE&T

Term 3, 2026

ELEC2134 : Circuits and Signals (PART A)

Topic 5: System Analysis (Bode Plot)

A Bode plot shows how the gain and phase of a system change with frequency, allowing us to understand how the system behaves at different frequencies.

- Convert transfer-function magnitude to dB.
- Sketch straight-line magnitude and phase Bode plots from factored H(s).
- Infer transfer-function structure from a Bode plot

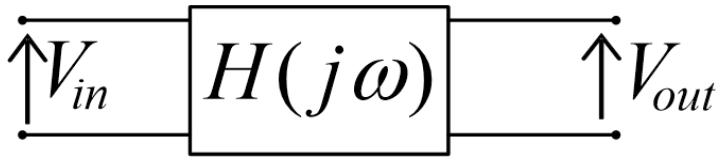
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Decibels (dB)

- The ratio of any two values of the same quantity (power, voltage, current) can be expressed in decibels (dB)
- For instance, we say that an amplifier has 10 dB power gain or a transmission line has a power loss of 7 dB (or gain -7 dB [ie loss]).
- If the gain/loss is 0 dB, the output magnitude equals the input magnitude.
- It is helpful to express the magnitudes of transfer functions in decibels.

- To convert a transfer-function magnitude to decibels, we multiply the common logarithm (base 10) of the transfer function magnitude by 20.



$$|H(j\omega)|_{dB} = 20\log_{10}\left|\frac{V_{out}(j\omega)}{V_{in}(j\omega)}\right|$$

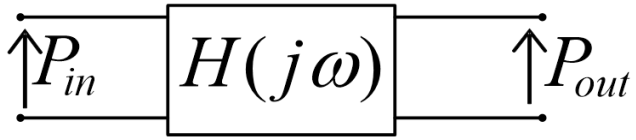
Voltage

Magnitude

Used for voltage transfer function

Decibels (dB)

- Ratios of powers are converted to decibels by taking 10 times the logarithm of the ratio.



$$dB = 10\log_{10}\left|\frac{P_{out}}{P_{in}}\right|$$

Power

Used for power transfer function

- To gain some feeling for the size of the decibel unit, recall some of the properties of the logarithm:
- Note: log 1 = 0, log 2 = 0.30103 & log 10 = 1

If $|H(j\omega)| = 1$, then $|H(j\omega)|_{dB} = 20\log_{10}1 = 0$ dB

If $|H(j\omega)| = 2$, then $|H(j\omega)|_{dB} = 20\log_{10}2 = 6$ dB ⟸ Voltage Ratio

If $|H(j\omega)| = 10$, then $|H(j\omega)|_{dB} = 20\log_{10}1 = 20$ dB

Note that the decibel equivalents are positive for magnitudes greater than unity, whereas the decibel equivalents are negative for magnitude less than unity.

An increase of $|H(j\omega)|$ by a factor of 10 corresponds to an increase in $|H(j\omega)|_{dB}$ by 20 dB.

- 1000 corresponds to 60 dB; $\Rightarrow dB = 20\log_{10}1000 = 60$
- 0.01 is represented as -40 dB; $\Rightarrow dB = 20\log_{10}0.01 = -40$ dB

$\lvert H\rvert$	dB
10	20
2	6.02
1	0
$1/\sqrt{2}$	−3.01
1/2	−6.02
0.1	−20

Decibels (dB)

Note: $\sqrt{2}$ = 3 dB and $\frac{1}{\sqrt{2}}$ = -3dB; $\boldsymbol{If\ H_{dB} = 20\log|H(j\omega)|\ \ then\ \ |H(j\omega)| = 10^{\frac{H_{dB}}{20}}}$

-3 dB corresponds to amplitude ratio $\frac{1}{\sqrt{2}}$, i.e. half power under equal impedance.

Note: $\mathrm{dB} = 10\log\left|\frac{P_{out}}{P_{in}}\right|$ ← Power ratio If $\left|\frac{P_{out}}{P_{in}}\right| = 10$ then $\mathrm{dB} = 10$

If $\left|\frac{P_{out}}{P_{in}}\right| = 1000000$ then $\mathrm{dB} = 60$ If $\left|\frac{P_{out}}{P_{in}}\right| = 100$ then $\mathrm{dB} = 20$

If $\left|\frac{P_{out}}{P_{in}}\right| = 1$ then $\mathrm{dB} = 0$

Exercise

Q1: Calculate $|H(j\omega)|_{dB}$ at ω=146 rad/s if:

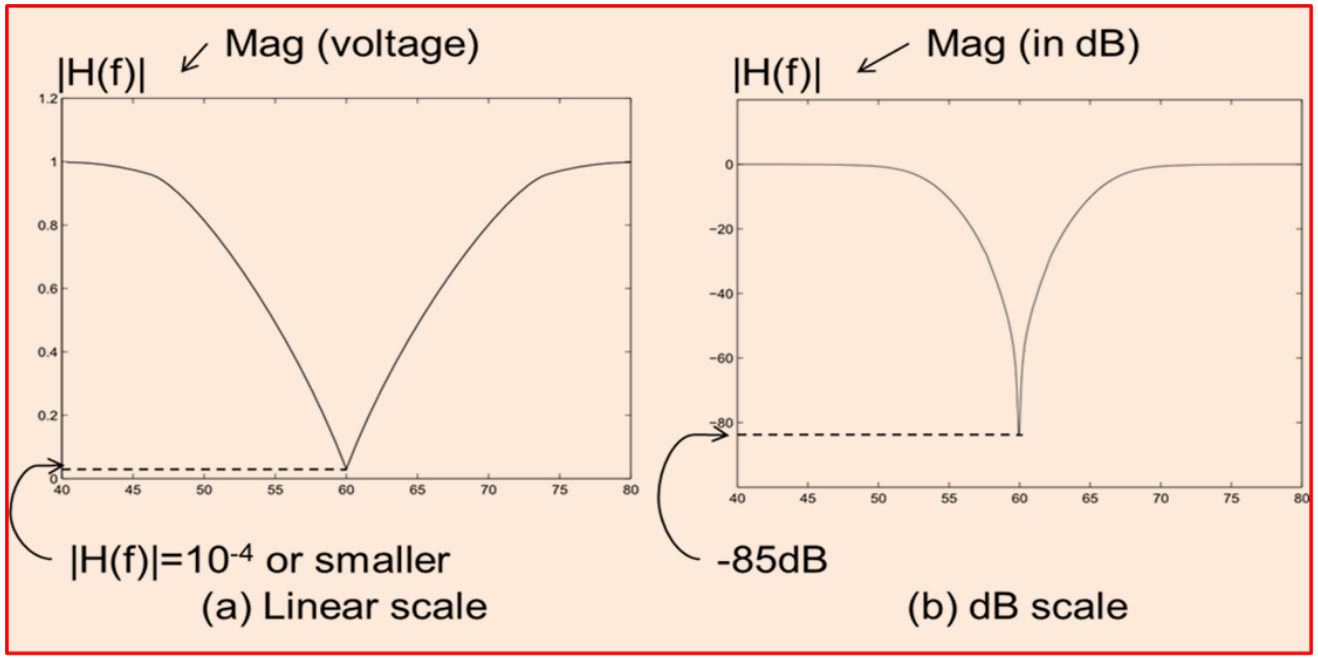
(a) H(s) = 20/(s+100) (b) H(s) = 20(s+100) (c) H(s) = 20s

Q2: Calculate |H(jω)| if H_{dB} equals

(a) 29.2 dB (b) -15.6 dB (c) -0.318 dB

Decibels (dB)

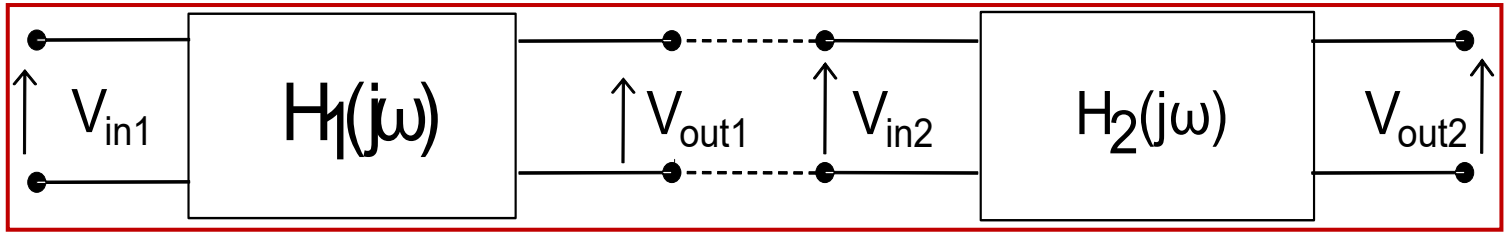
- A magnitude of a filter transfer function is shown in Fig (a) and (b)



✓ When the magnitude is converted to decibels, passband magnitude is approximately unity (0 dB) and that at 60 Hz the magnitude is sufficiently small (less than -80 dB)

✓ One advantage of converting transfer function magnitudes to decibels before plotting is that very small and very large magnitude can be displayed clearly on a single plot.

Cascade Connection



$$H_1(j\omega) = \frac{V_{out1}}{V_{in1}}; \quad H_2(j\omega) = \frac{V_{out2}}{V_{in2}}$$

But $V_{out1} = V_{in2}$

$$H(j\omega) = \frac{V_{out2}}{V_{in1}} = \frac{V_{out2}}{V_{in2}} \times \frac{V_{out1}}{V_{in1}}$$

V_{in2}

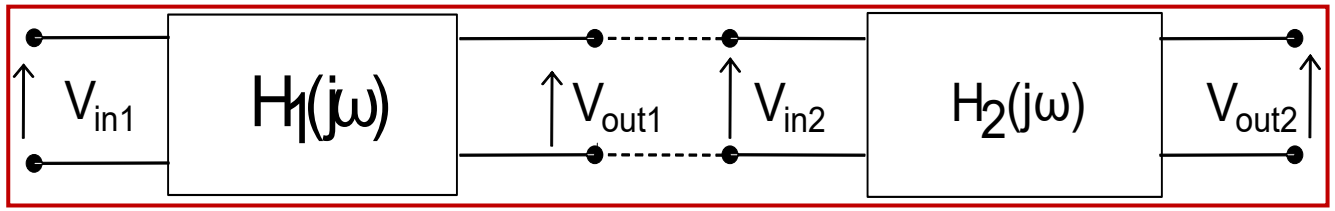
$$= H_2(j\omega) \times H_1(j\omega)$$

$$\therefore H(j\omega) = H_1(j\omega)H_2(j\omega)$$

Overall Transfer function

When we connect the output terminals of one two-port circuit to the input terminals of another two-port circuit, we say that we have a cascade connection

Cascade Connection



$$H_1(j\omega) = \frac{V_{out1}}{V_{in1}}; \quad H_2(j\omega) = \frac{V_{out2}}{V_{in2}}$$

But $V_{out1} = V_{in2}$

$$H(j\omega) = \frac{V_{out2}}{V_{in1}} = \frac{V_{out2}}{V_{in2}} \times \frac{V_{out1}}{V_{in1}}$$

$$= H_2(j\omega) \times H_1(j\omega)$$

$$\therefore H(j\omega) = H_1(j\omega)H_2(j\omega)$$

Overall Transfer function

- When we connect the output terminals of one two-port circuit to the input terminals of another two-port circuit, we say that we have a cascade connection
- Thus, the transfer function of the cascade connection is the product of the transfer function of the individual two-port circuit/system
- This fact can be extended to three or more two-ports connected in cascade
- We have

$$H(j\omega) = H_1(j\omega) \times H_2(j\omega)$$

$$|H(j\omega)| = |H_1(j\omega)||H_2(j\omega)|$$

- By expressing in decibels, we get

$$20\log|H(j\omega)| = 20\log|H_1(j\omega)| + 20\log|H_2(j\omega)|$$

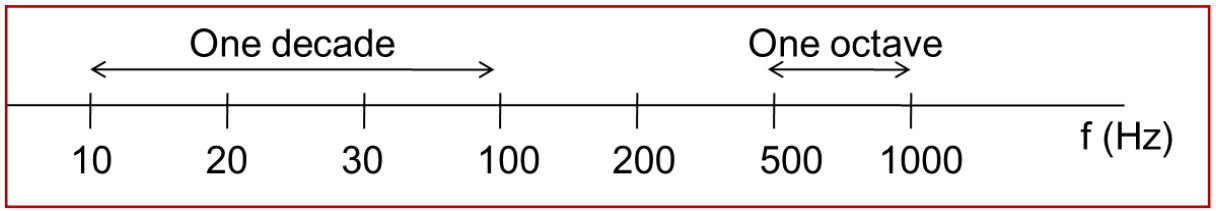
$$|H(j\omega)|_{dB} = |H_1(j\omega)|_{dB} + |H_2(j\omega)|_{dB}$$

Products become sums in dB

Thus in decibels, the individual transfer function magnitudes are added to find the overall transfer function magnitude for a cascade connection

Logarithmic Frequency Scales

- We often use a logarithmic scale for frequency when plotting transfer functions.
- For example, a logarithmic frequency scale is shown below:



- A **decade** is a range of frequencies for which the ratio of the highest frequency to the lowest is 10.
- The frequency range from 2 to 20 Hz is one decade.
- Similarly, the range from 50 to 5000 Hz is two decades (50 to 500 Hz is one decade, and 500 to 5000 Hz is another decade).
- An **octave** is a two-to-one change in frequency. For example, 10 to 20 Hz is one octave. The range 2 to 16 Hz is three octaves.

- Suppose that we have two frequencies f_1 and f_2 to which $f_2 > f_1$. The number of decades between f_1 and f_2 is given by,

$$\text{Number of decades} = \log_{10}\frac{f_2}{f_1}$$

- The number of octaves between two frequencies is:

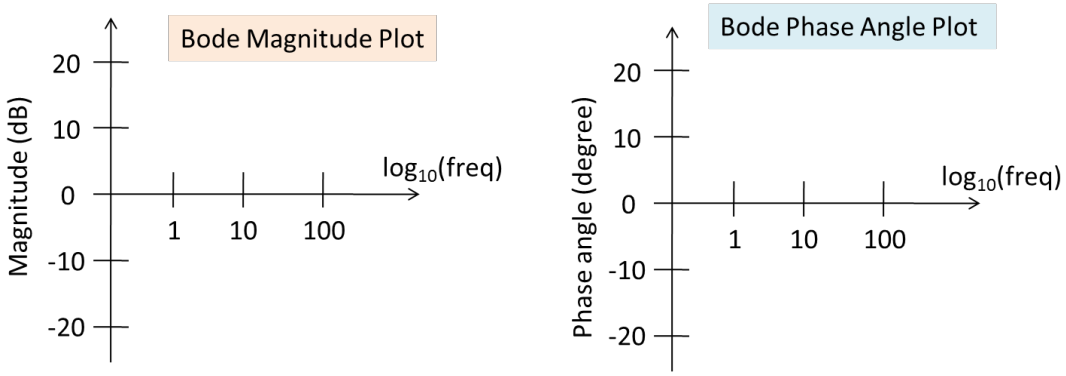
$$\text{Number of octaves} = \log_2\frac{f_2}{f_1} = \frac{\log_{10}\frac{f_2}{f_1}}{\log_{10}2}$$

- The advantage of a logarithmic frequency scale compared with a linear scale is that the variations in the magnitude or phase of a transfer function for a low range of frequency such as 10 to 20 Hz as well as the variations in high ranges such as 10 to 20 MHz can be clearly shown on a single plot.

Note: With linear scale, either the low range would be severely compressed or the high range would be off scale]

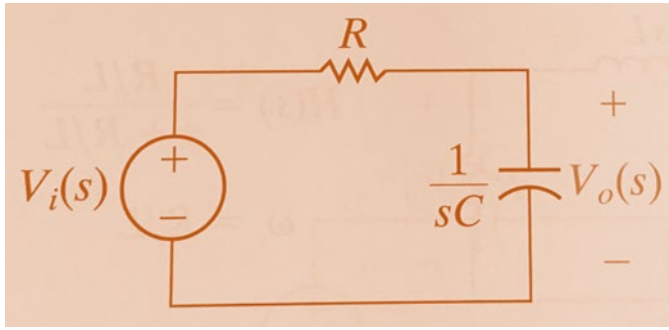
Magnitude and Phase Plots

- A Bode Plot shows both the magnitude and phase curves using a logarithmic frequency scale for the abscissa (normally x-axis) and magnitude itself is also shown in logarithmic unit called dB (for the magnitude plot) and degrees for the phase plot.



- The Bode Plot is particularly useful for displaying transfer functions.
- The Bode Plots can often be closely approximated by straight-line segments, so that they are relatively easy to draw.

✓Consider the first-order lowpass filter transfer function:



$$H(j\omega) = \frac{1}{1 + jRC\omega} = \frac{1}{1 + j\dfrac{\omega}{\omega_C}} \qquad Let \quad \omega_c = \frac{1}{RC}$$

$$|H(j\omega)| = \frac{1}{\sqrt{1 + \left(\dfrac{\omega}{\omega_C}\right)^2}}$$

Magnitude

- To convert the magnitude to decibels, we take 20 times the logarithm of the magnitude $|H(j\omega)|$:

Magnitude and Phase Plots

$$|H(j\omega)| \text{ in dB} = 20\log\frac{1}{\sqrt{1+\left(\frac{\omega}{\omega_C}\right)^2}} = 20\log 1 - 20\log\sqrt{1+\left(\frac{\omega}{\omega_C}\right)^2}$$

$$= 0 - 20\log\left[1+\left(\frac{\omega}{\omega_C}\right)^2\right]^{1/2} = -10\log\left[1+\left(\frac{\omega}{\omega_C}\right)^2\right]$$

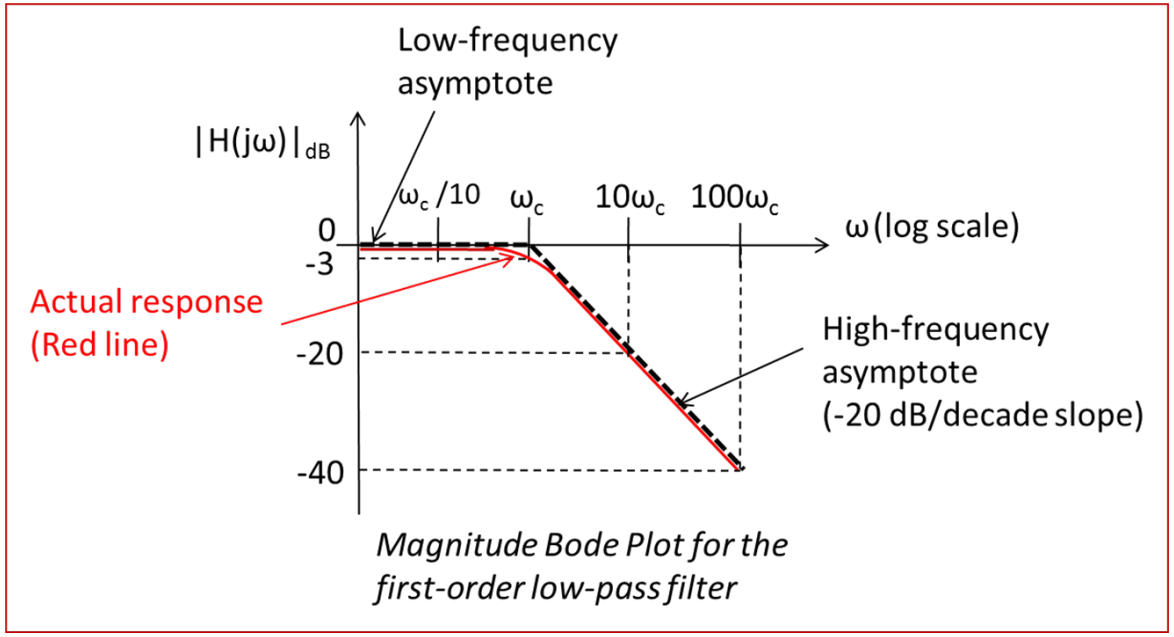
$$|H(j\omega)| \text{ in dB} = -10\log\left[1+\left(\frac{\omega}{\omega_C}\right)^2\right]$$

$$|H(j\omega)| \approx 1$$

- If $\omega_c >>> \omega$ then $|H(j\omega)| \approx -10\log(1) = 0$. Thus, for low frequencies, the transfer function magnitude is approximated by the horizontal straight line shown in figure below, labelled as the low-frequency asymptote.

✓On the other hand $\omega_c << \omega$ (i.e. $\omega/\omega_c >> 1$)

$$|H(j\omega)|_{dB} \approx -10\log\left(\frac{\omega}{\omega_C}\right)^2 = -20\log\left(\frac{\omega}{\omega_C}\right)$$

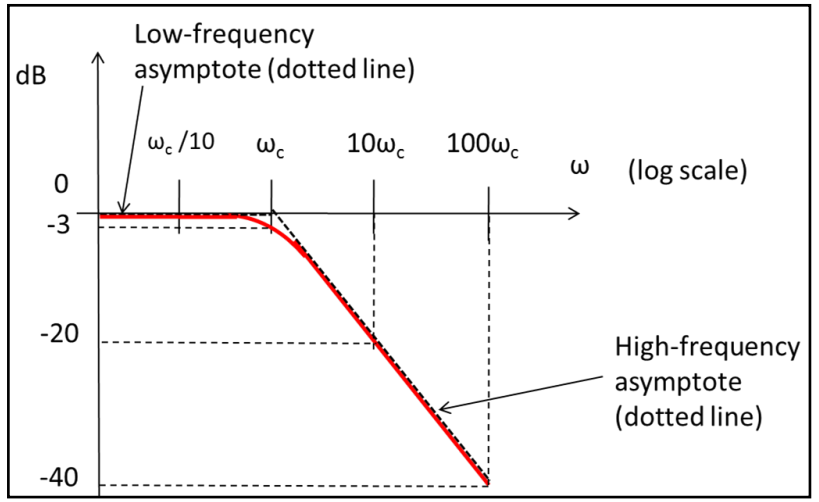


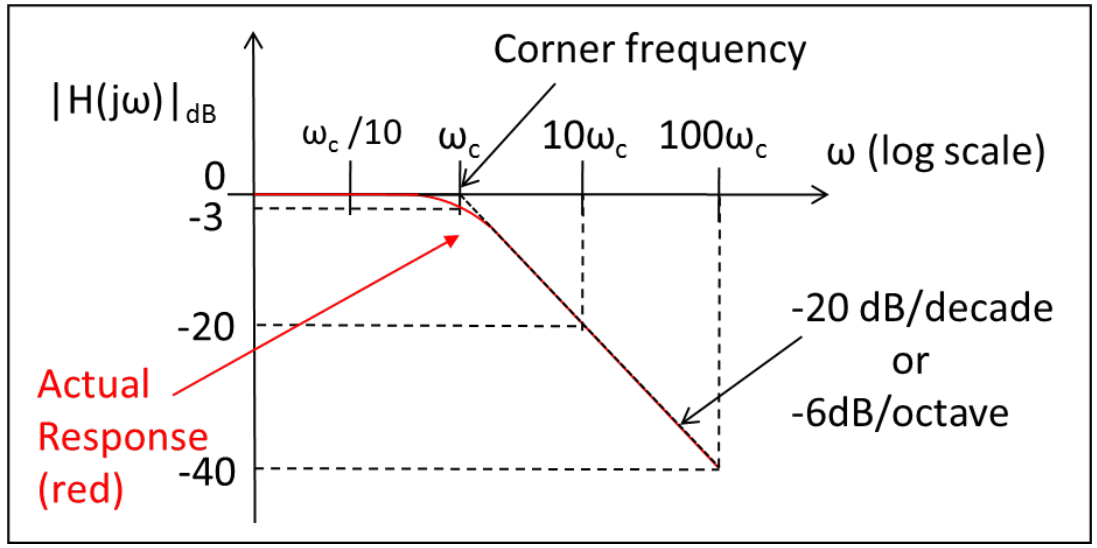
Magnitude Bode Plot for the first-order low-pass filter

Magnitude and Phase Plots

- Evaluating for various values of ω, we obtain the results shown below:

ω	$\lvert H(j\omega)\rvert_{dB} = -10\log\left[1+\left(\frac{\omega}{\omega_c}\right)^2\right]$
ω_c	0
$2\omega_c$	-6
$10\omega_c$	-20
$100\omega_c$	-40
$1000\omega_c$	-60





Thus, the asymptotes are in error by only 3 dB at corner frequency. The actual curve (red) for $|H(j\omega)|_{dB}$ is shown above.

- Plotting the above table values result in a straight line sloping downwards and is labelled as high-frequency asymptote;
- Notice that the two straight line asymptotes intersect at the half-power frequency ωc.
- For this reason, ωc is also known as the corner (cutoff) frequency.
- **Note:** The slope of the high frequency asymptote is -20 dB/decade of frequency (This slope can also be stated as -6 dB per octave).
- If we evaluate $|H(j\omega)|_{dB}$ at ω= ωc, we find that

$$\left|H(j\omega)\right|_{\omega=\omega_c} = -10\log\left(1+\frac{\omega_c}{\omega_c}\right) = -3dB$$

Phase Plot

$$H(j\omega) = \frac{1}{1 + jRC\omega} = \frac{1}{1 + j\dfrac{\omega}{\omega_C}} \qquad Let \quad \omega_c = \frac{1}{RC}$$

- The phase is given by

$$\angle H(j\omega) = -\arctan\left(\frac{\omega}{\omega_c}\right)$$

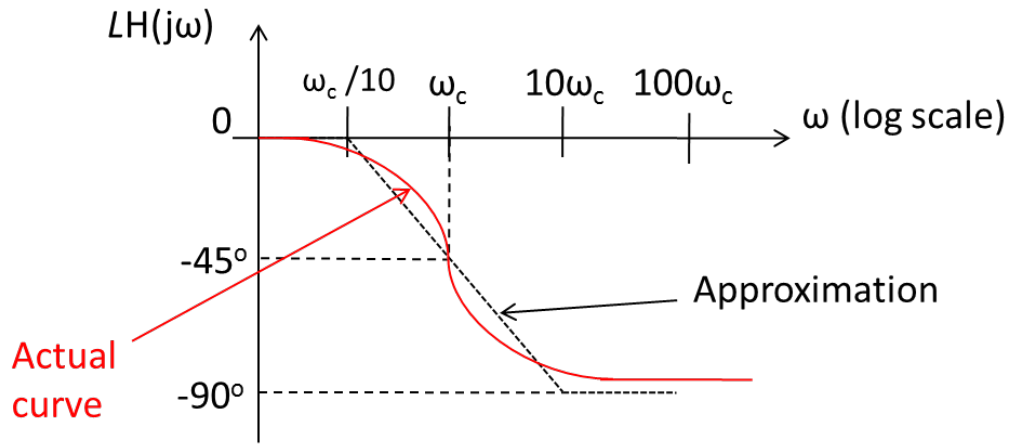
- Evaluating $\angle H(j\omega)$ when $\omega_c >> \omega$ i.e. $\omega/\omega_c << 1$, we find that the phase approaches zero at low frequencies.

- When $\omega = \omega_c$ (at corner frequency)

$$\angle H(j\omega) = -\arctan\left(\frac{\omega_c}{\omega_c}\right) = -\arctan(1) = -45^\circ$$

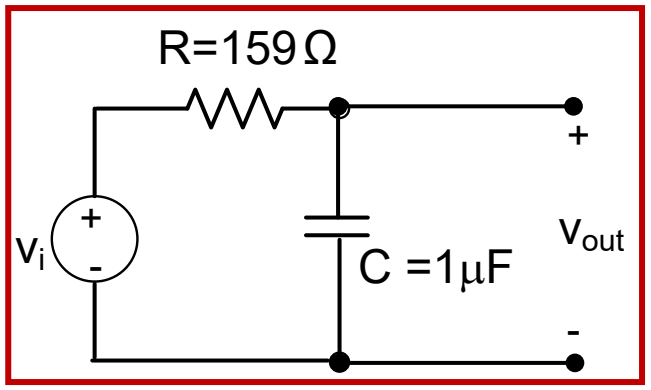
- When $\omega_c << \omega$ i.e. $\omega/\omega_c >> 1$, then $\angle H(j\omega)$ approaches to -90° at high frequencies

- Figure below shows a plot of phase versus frequencies for the first order lowpass filter.
- The phase plot can be approximated by the following straight-line segments:
  - A horizontal line at zero for $\omega < \omega_c/10$
  - A sloping line from zero phase at $\omega_c/10$ to -90° at $10\omega_c$.
  - A horizontal line at -90° for $\omega > 10\omega_c$.



Example 1

- Sketch the approximate straight-line Bode magnitude and phase plots to scale for the circuit shown below:

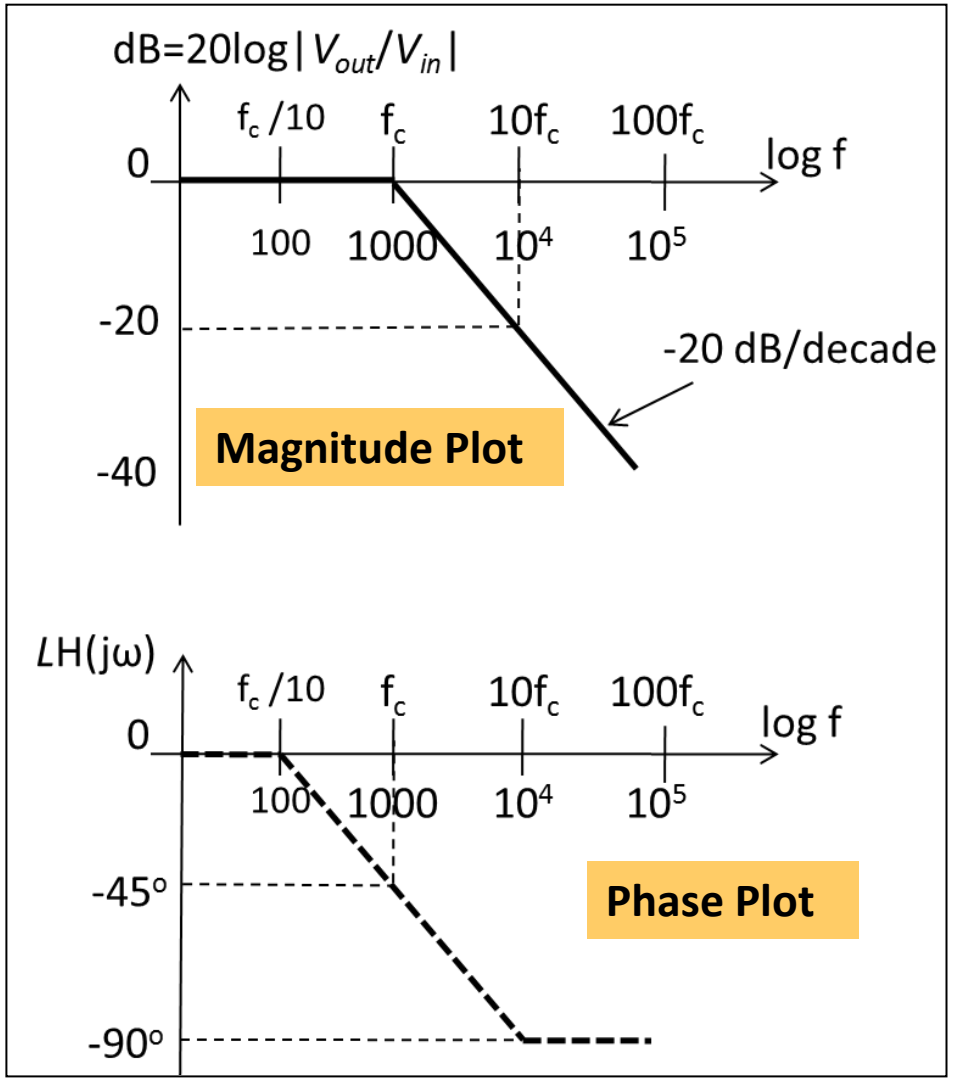


$$\frac{V_{out}}{V_i} = \frac{1}{1 + j\omega RC}$$

$$CR = 1\times10^{-6}\times159 = 159\times10^{-6}$$

$$\omega_c = \frac{1}{CR} = \frac{1}{159\times10^{-6}}$$

$$f_c = \frac{1}{2\pi\times159\times10^{-6}} = 1000\,Hz$$



Example 2

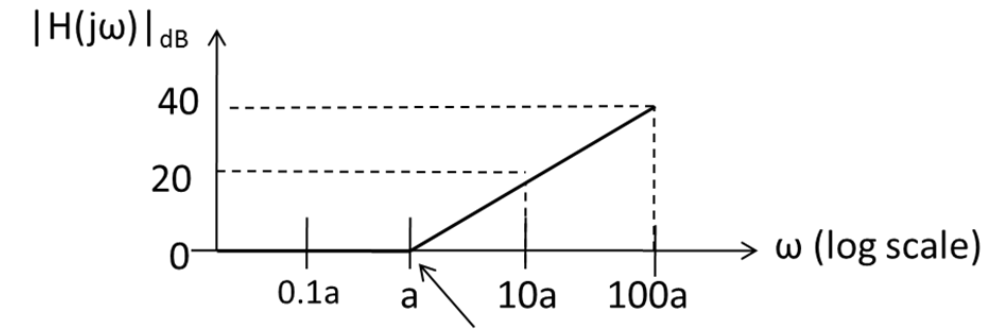
- Sketch the approximate straight-line Bode magnitude plot for $H(s) = 1 + \frac{s}{a}$

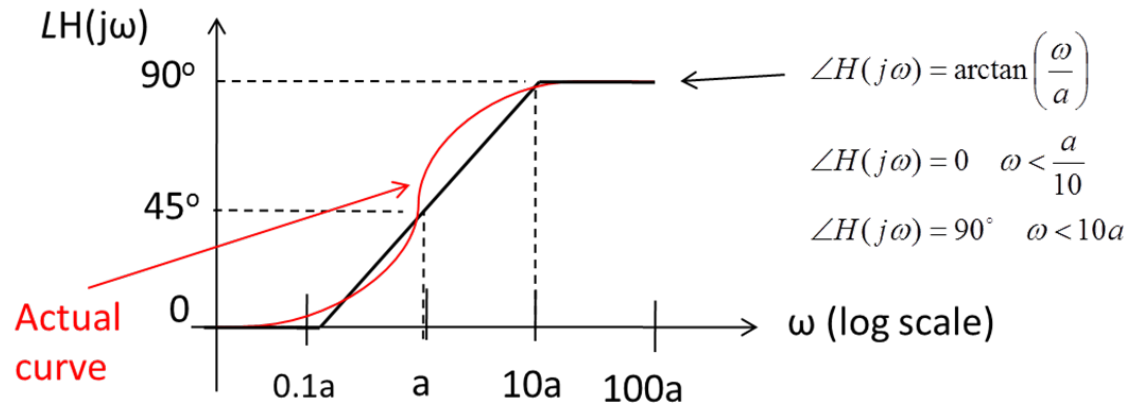
$$H(j\omega) = 1 + \frac{j\omega}{a} \quad \Rightarrow \quad |H(j\omega)| = \sqrt{1 + \frac{\omega^2}{a^2}}$$

$$|H(j\omega)|_{dB} = 20\log\left(1 + \frac{\omega^2}{a^2}\right)^{1/2}$$

$$\text{When } \omega << a, |H(j\omega)|_{dB} = 20\log(1) = 0$$
$$\text{When } \omega >> a, |H(j\omega)|_{dB} = 20\log(\frac{\omega}{a})$$

- ✓ At ω=a, $|H(j\omega)|_{dB}$ = 0; At ω=10a, $|H(j\omega)|_{dB}$ = 20 dB; At ω=100a, $|H(j\omega)|_{dB}$ = 40 dB;
- ✓ Thus, the value of $|H(j\omega)|_{dB}$ increases 20 dB for every 10-fold increase in frequency.
- ✓ The asymptote therefore has a slope of 20 dB/decade.
- ✓ Since $|H(j\omega)|_{dB}$ increases by 6 dB when ω doubles, an alternate value for the slope is 6 dB/octave





Example 3

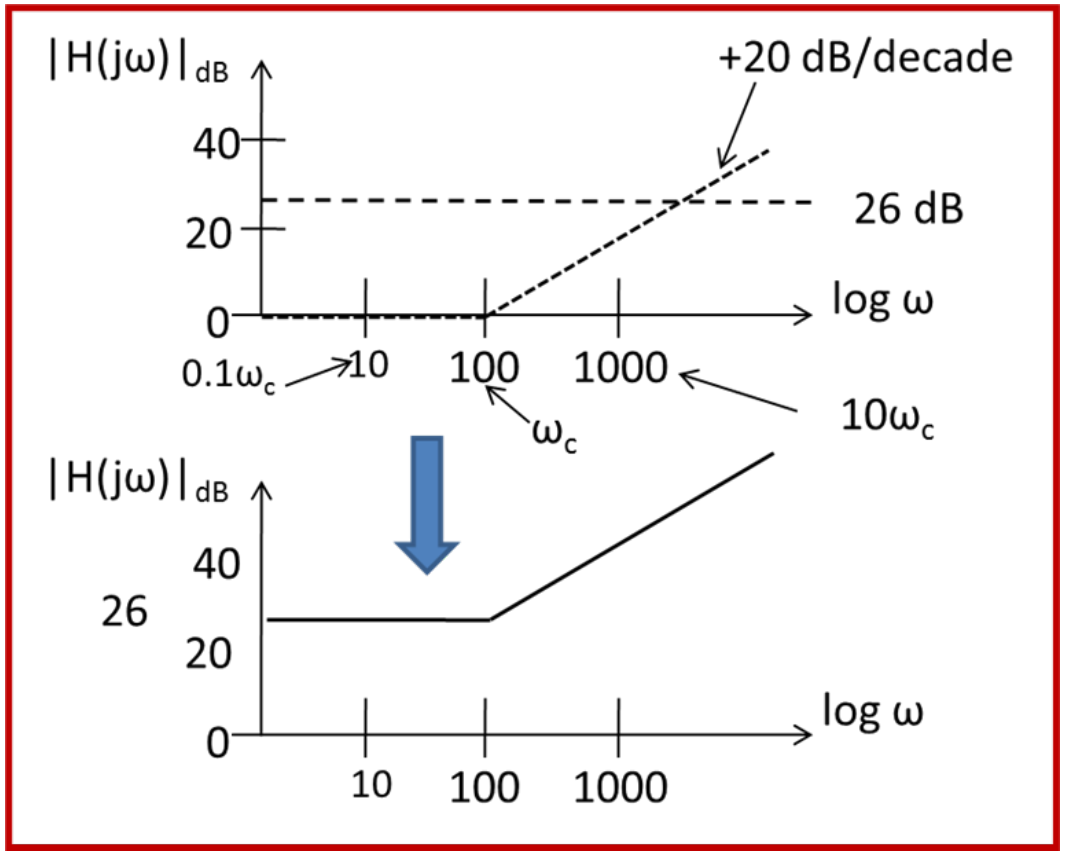
- Obtain the Bode plot for the transfer function H(s) = 20 + 0.2s
- Putting this in standard form, we obtain

$$H(s) = 20\left(1 + \frac{s}{100}\right) \Rightarrow H(j\omega) = 20\left(1 + \frac{j\omega}{100}\right) \quad Let\ \omega_c = 100$$

$$|H(j\omega)| = 20\left[1 + \left(\frac{\omega}{100}\right)^2\right]^{1/2}$$

$$|H(j\omega)|_{dB} = 20\log 20 + 10\log\left[1 + \left(\frac{\omega}{100}\right)^2\right]$$

$$= 26dB + 10\log\left[1 + \left(\frac{\omega}{100}\right)^2\right]$$



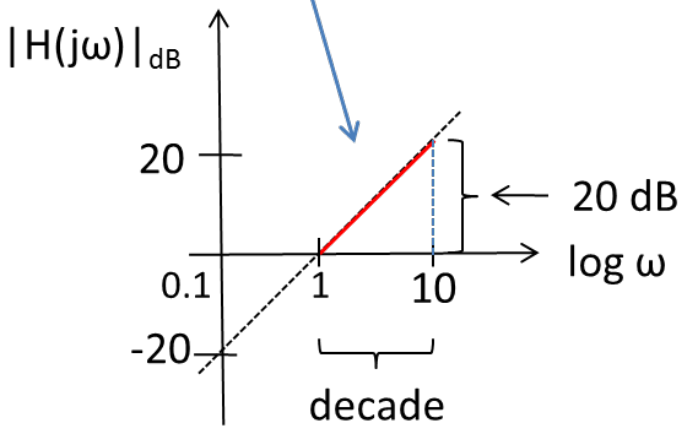
Example 4

- Bode plots of a differentiator and an Integrator

Diffentiator

$$H(s) = s$$
$$H(j\omega) = j\omega$$
$$|H(j\omega)| = \omega$$
$$|H(j\omega)|_{dB} = 20\log\omega$$

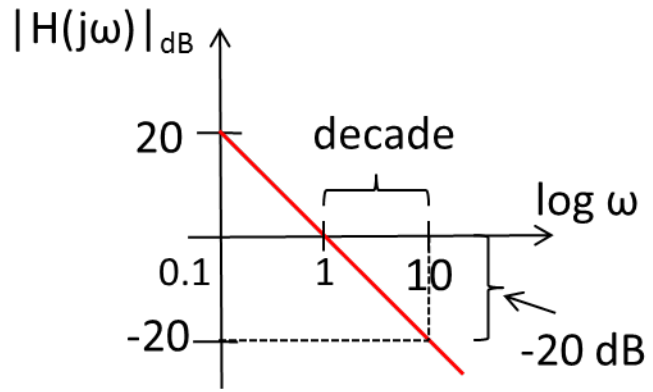
20 dB/decade slope



Mirror image of the bode plots with respect to ω-axis

Integrator

$$H(s) = \frac{1}{s}$$
$$H(j\omega) = \frac{1}{j\omega}$$
$$|H(j\omega)| = \frac{1}{\omega}$$
$$|H(j\omega)|_{dB} = 20\log\frac{1}{\omega} = -20\log\omega$$



Both plots are infinitely long straight line passing through 0 dB at ω = 1 and having slope ±20 dB/decade.

Example 5 – Magnitude Plot

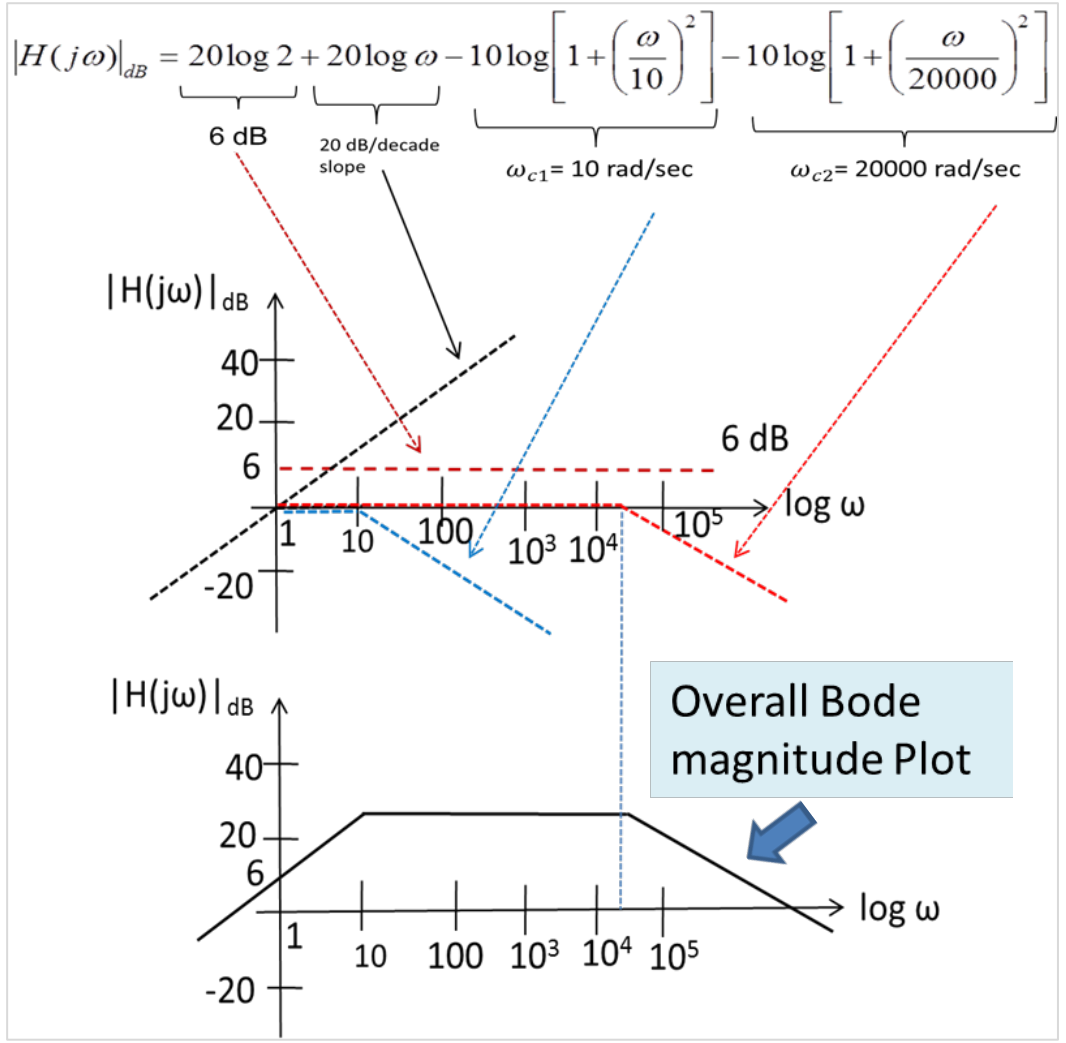
- Construct a Bode magnitude plot for $H(s) = \dfrac{-400000s}{(s+10)(s+20000)}$

$$H(s) = \frac{-2s}{\left(1+\frac{s}{10}\right)\left(1+\frac{s}{20000}\right)} \Rightarrow H(j\omega) = \frac{-2j\omega}{\left(1+\frac{j\omega}{10}\right)\left(1+\frac{j\omega}{20000}\right)}$$

ω_{c1} → 10 ← 20000 ← ω_{c2}

$$|H(j\omega)| = \frac{2\omega}{\sqrt{1+\left(\frac{\omega}{10}\right)^2}} \frac{1}{\sqrt{1+\left(\frac{\omega}{20000}\right)^2}}$$

$$|H(j\omega)|_{dB} = \underbrace{20\log 2}_{6\text{ dB}} + \underbrace{20\log\omega}_{20\text{ dB/decade slope}} - \underbrace{10\log\left[1+\left(\frac{\omega}{10}\right)^2\right]}_{\omega_{c1}=10\text{ rad/sec}} - \underbrace{10\log\left[1+\left(\frac{\omega}{20000}\right)^2\right]}_{\omega_{c2}=20000\text{ rad/sec}}$$



Example 5 – Phase Plot

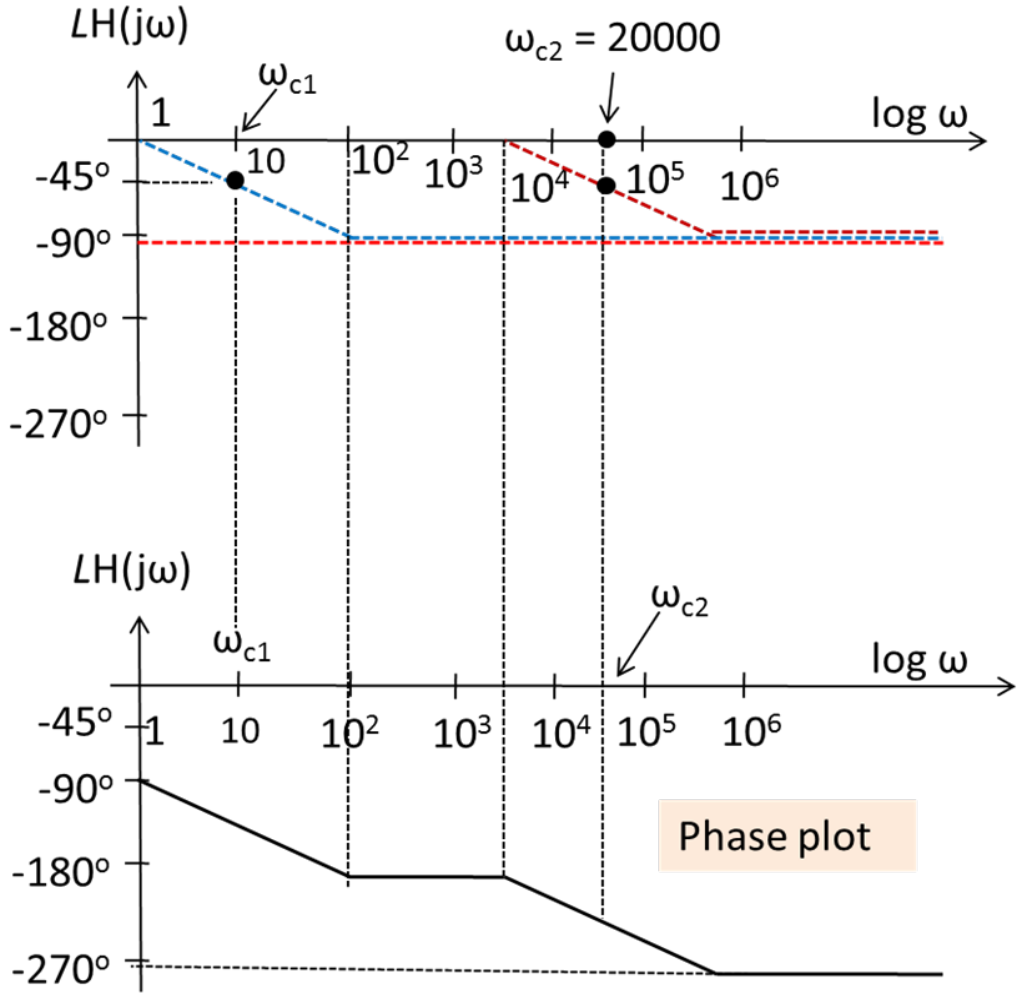
- Construct a Bode phase plot for $H(s) = \dfrac{-400000s}{(s+10)(s+20000)}$

$$H(s) = \frac{-2s}{\left(1+\frac{s}{10}\right)\left(1+\frac{s}{20000}\right)}$$

$$H(j\omega) = \frac{-2j\omega}{\left(1+\frac{j\omega}{10}\right)\left(1+\frac{j\omega}{20000}\right)}$$

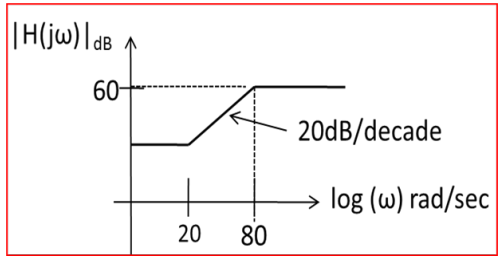
$$H(j\omega) = \frac{-j(2\omega)}{\left(1+\frac{j\omega}{10}\right)\left(1+\frac{j\omega}{20000}\right)}$$

The angle of the numerator is a constant, -90° angle



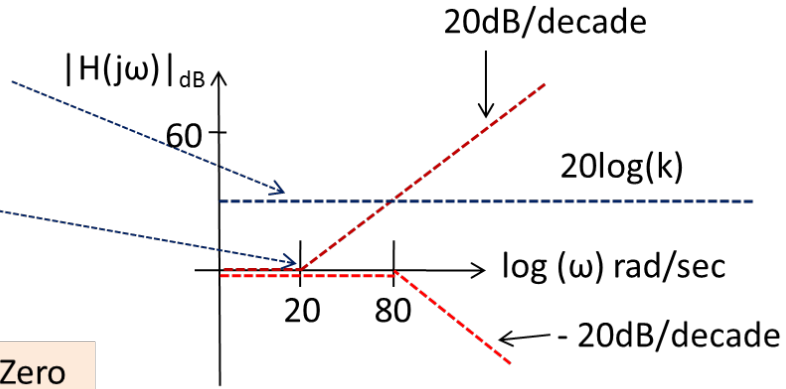
Example 6

- The diagram on the right is a straight line approximation to the Bode magnitude for a particular system. Determine the transfer function in Bode form that could have been this magnitude response.



Solution

✓ There are three terms
- A constant term
- A zero at 20
- A pole at 80



$$\therefore H(j\omega) = k\frac{1+\frac{j\omega}{20}}{1+\frac{j\omega}{80}}$$

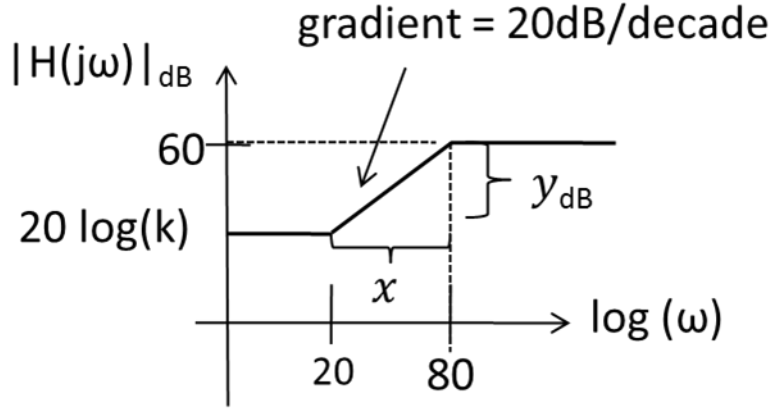
Zero

Pole

Constant

$$\therefore H(j\omega) = 250\frac{1+\frac{j\omega}{20}}{1+\frac{j\omega}{80}}$$

How to find K?



Number of decades between the corner frequencies
= log(80) – log(20) = 0.602

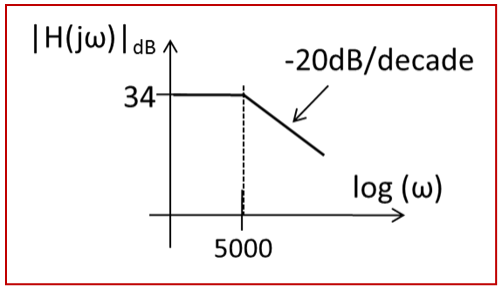
$\frac{y_{dB}}{x} = 20;\ \therefore y_{dB} = 20 \times 0.602$

$\therefore$ 20 log(k) = 60dB - y_{dB} = 47.96 dB

$log_{10}(k) = \frac{47.96}{20};\ \therefore k = 10^{\frac{47.96}{20}} = 250$

Example 7

- The diagram on the right is a straight line approximation to the Bode magnitude for a particular system. Determine the transfer function in Bode form that could have been this magnitude response.



Solution

$$\therefore |H(j\omega)| = \frac{k}{1+\frac{j\omega}{5000}}$$

$$\therefore 20\log|H(j\omega)| = 20\log k - 20\log\sqrt{1+\left(\frac{\omega}{5000}\right)^2}$$

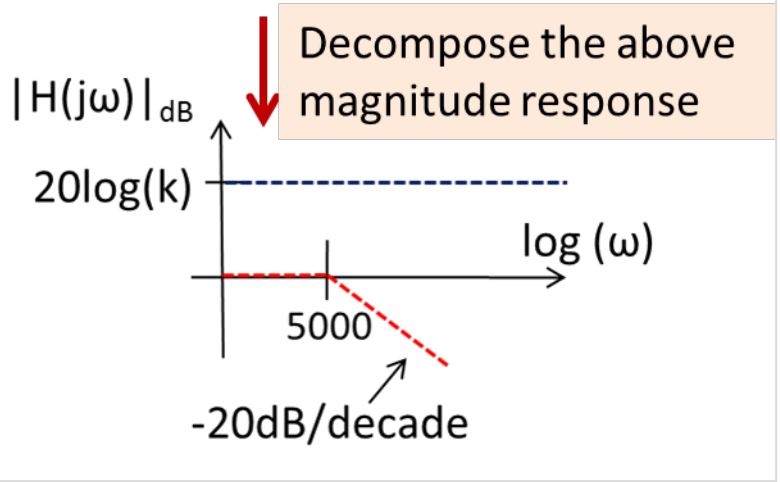
$$= 20\log k - 10\log\left(1+\left(\frac{\omega}{5000}\right)^2\right)$$

$$20\log_{10} k = 34$$

$$k = 10^{\frac{34}{20}} = 50.12$$

$$\therefore H(j\omega) = \frac{50.12}{1+\frac{j\omega}{5000}} \quad or \quad H(s) = \frac{50.12\times 5000}{5000+s}$$

Decompose the above magnitude response



Example 8

- A transfer function G(s) is given by $G(s) = \dfrac{10s^2}{(s+50)^2}$

(a) Rewrite G(s) in the Bode form.
(b) Determine an expression for the magnitude response $20\log_{10}|G(j\omega)|$
(c) Determine an expression for the phase of response $G(j\omega)$.
(d) Draw straight line approximations of Bode log magnitude and phase plots of $G(j\omega)$

Solution

(a)

$$G(j\omega) = \frac{10(j\omega)^2}{(j\omega+50)^2} = \frac{10(j\omega)^2}{2500\left(1+\frac{j\omega}{50}\right)^2} = \frac{1}{250}\frac{(j\omega)^2}{\left(1+\frac{j\omega}{50}\right)^2}$$

(b)

$$\therefore |G(j\omega)| = \left|\frac{1}{250}\right| \frac{|j\omega|^2}{\left|\left(1+\frac{j\omega}{50}\right)^2\right|}$$

$$20\log|G(j\omega)| = 20\log\left(\frac{1}{250}\right) + 20\log|j\omega|^2 - 20\log\left\{\left|1+\frac{j\omega}{50}\right|\right\}^2$$

$$20\log|G(j\omega)| = -47.96 + 40\log|\omega| - 40\log\left\{1+\left(\frac{\omega}{50}\right)^2\right\}$$

(c)

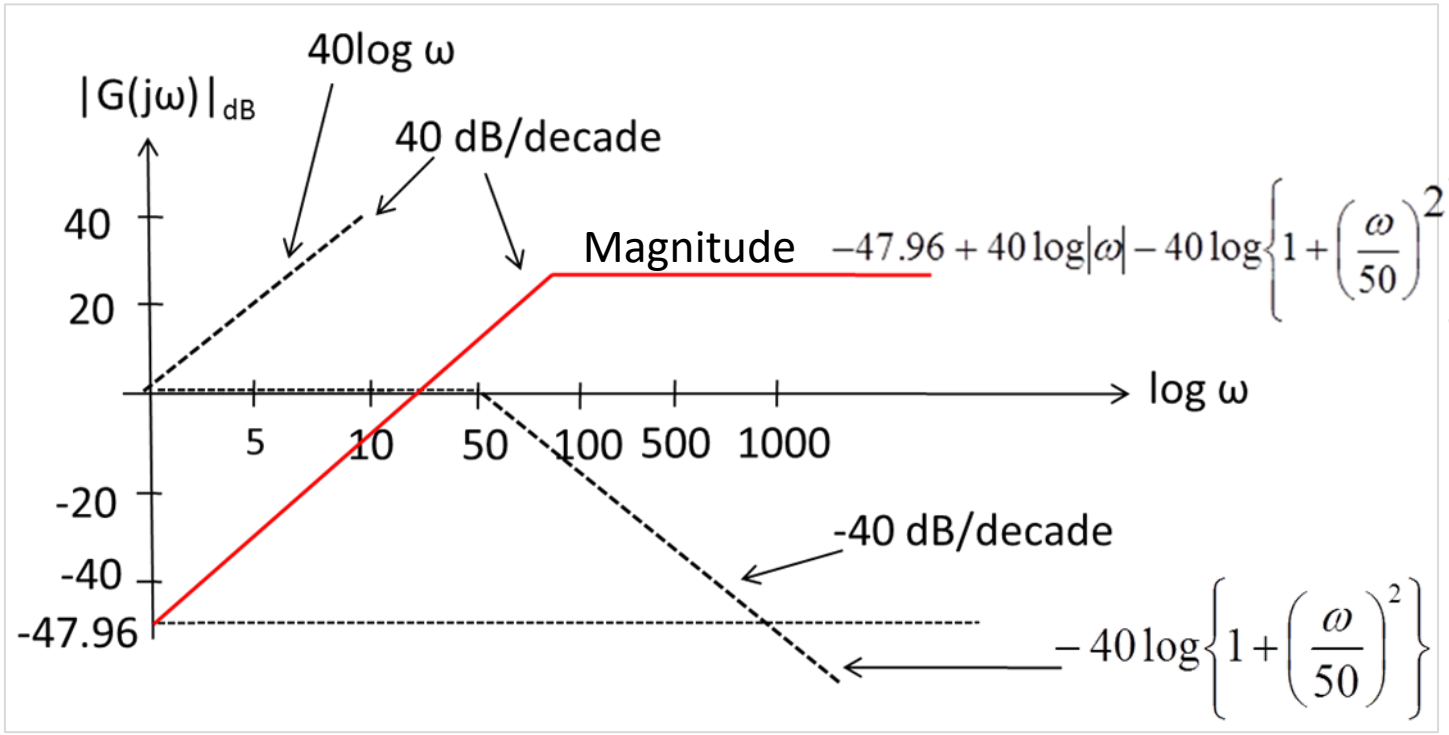
$$G(j\omega) = \frac{1}{250}\frac{(j\omega)^2}{\left(1+\frac{j\omega}{50}\right)^2}$$

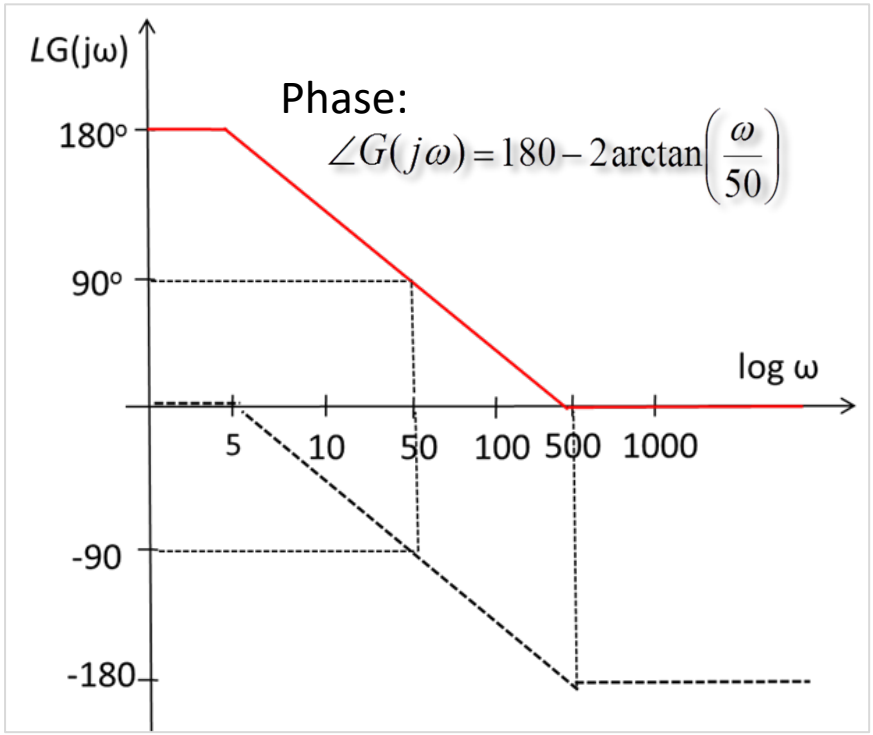
$$\angle G(j\omega) = \angle\frac{1}{250} + \angle j\omega + \angle j\omega - \angle(1+\frac{j\omega}{50}) - \angle(1+\frac{j\omega}{50})$$

$$= 0 + 90^\circ + 90^\circ - \arctan\left(\frac{\omega}{50}\right)\times 2$$

$$= 180 - 2\arctan\left(\frac{\omega}{50}\right)$$

Example 8





Exercise

Sketch Bode plot for each of the following transfer functions:

(a) $H(s)=1$ (b) $H(s)=s^2$ (c) $H(s)=s^3$ (d) $H(s)=\frac{1}{s^3}$ (e) $\mathrm{H(s)}=\frac{1}{10s+1}$

(f) $\mathrm{H(s)}=\frac{10}{s+10}$ (g) $\mathrm{H(s)}=10s+1$ (h) $\mathrm{H(s)}=\frac{s}{5}+1$ (i) $\mathrm{H(s)}=\frac{4}{s^2+2s+4}$

Pole vs Zero Intuition

Pole

- tends to reduce high-frequency magnitude
- contributes negative phase

$$\text{pole} \Rightarrow -20\ \text{dB/dec and } -90^\circ$$

Zero

- tends to increase high-frequency magnitude
- contributes positive phase

$$\text{zero} \Rightarrow +20\ \text{dB/dec and } +90^\circ$$

Repeated poles/zeros

If a pole is repeated twice:

$$\frac{1}{(1+s/\omega_p)^2}$$

then slope change:

$$\boxed{-40\ \text{dB/dec}}$$

and phase approaches:

$$\boxed{-180^\circ}.$$

Similarly double zero:

$$+40\ \text{dB/dec}, \quad +180^\circ.$$

Factor	Magnitude slope	Phase
K	0 dB/dec	0° if $K > 0$
s	$+20$ dB/dec	$+90^\circ$
$1/s$	-20 dB/dec	-90°
$1 + s/\omega_z$	adds $+20$ dB/dec after ω_z	$0 \to +90^\circ$
$1/(1 + s/\omega_p)$	adds -20 dB/dec after ω_p	$0 \to -90^\circ$

$$\text{Laplace} \;/\; H(s) \to \text{poles, zeros, stability and } H(j\omega) \to \text{Bode magnitude and phase}$$